Analyzing Human Motion for Predicting Penalty Direction in Football

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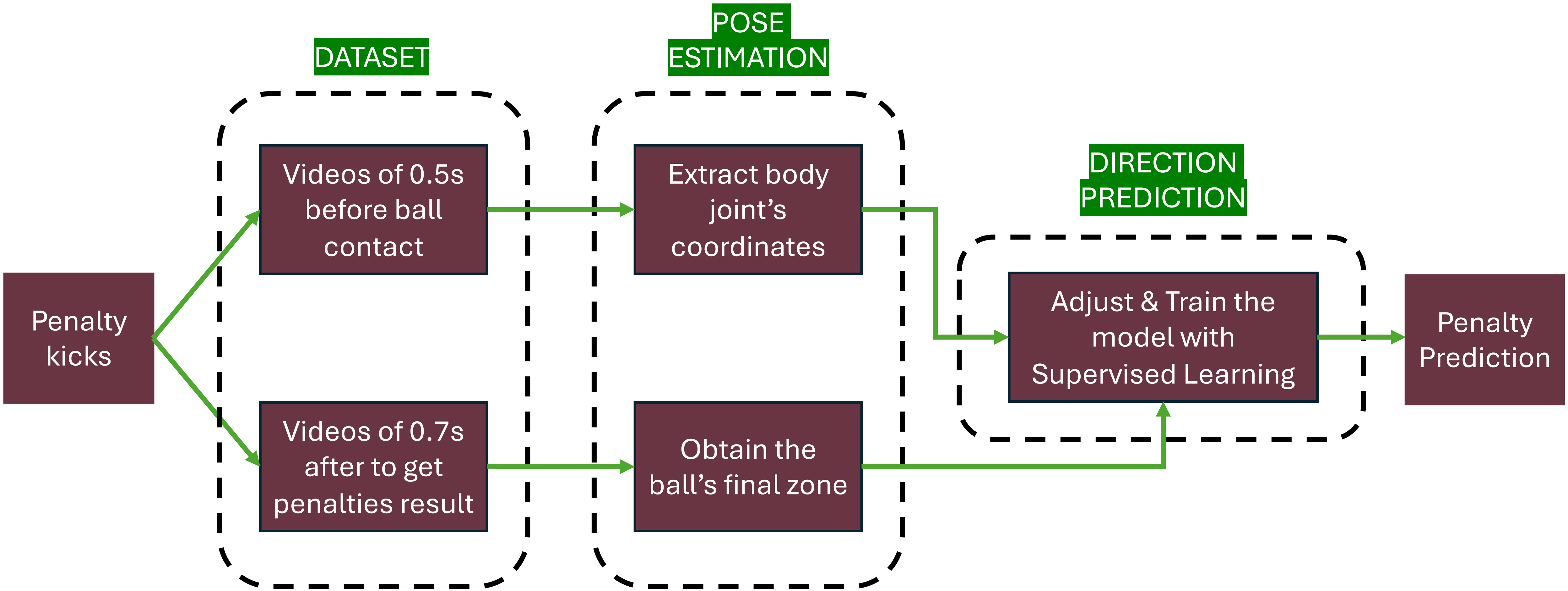
- MA4 – Robotics
- Semester project of 8 ECTS
- Supervised by Saeed Saadatnejad

Project

- **Dataset**
  - Made up of :
    - Penalty kick before ball contact
    - Penalty result
  - Large (min > 1'000 penalty kicks) → penalties from fifa games & automatic process.
- **Human body pose estimation**
  - Extract the body joints coordinates of the kicker
- **Train a transformer-based model with supervised learning**
  - Predict the direction of penalty kicks.


DATASET
POSE ESTIMATION
DIRECTION PREDICTION
Penalty kicks
Videos of 0.5s before ball contact
Videos of 0.7s after to get penalties result
Extract body joint’s coordinates
Obtain the ball’s final zone
Adjust & Train the model with Supervised Learning
Penalty Prediction

Previous work

- **Master Thesis project from June 2023 in KU Leuven (found in a Github repository)**
  - Penalty kick Analysis: Visual Recognition, Pose Estimation and LSTM.
- **Dataset made of actual penalties in Youtube**
- **Yolov7 for ball/player detection**
- **LSTM prediction**
- **Limitations**
  - Manual selection of penalties
    → small dataset (792 videos)
  - 3 zones
    → Left, Center & Right
  - Increase from 33.8% to 41.6%
    → improvement of only 7.8%

What I have I done

- **Dataset**
  - Apply detection (Yolov9) on the ball to obtain :
    - Penalty kick before ball contact, 0.5s videos ~ 30 frames.
    - Result of the penalty in 6 zones, with a kalman filter on the ball.
  - Processed 20 videos ~ 1’200 penalty kicks.
  - Difficulties encoutered :
    - Multiple outliers
    - Poor detection of the ball due tozo noisy background
- **Human body pose estimation**
  - Start to apply a pre-trained model from OpenPifPaf to obtain the player’s body joints coordinates.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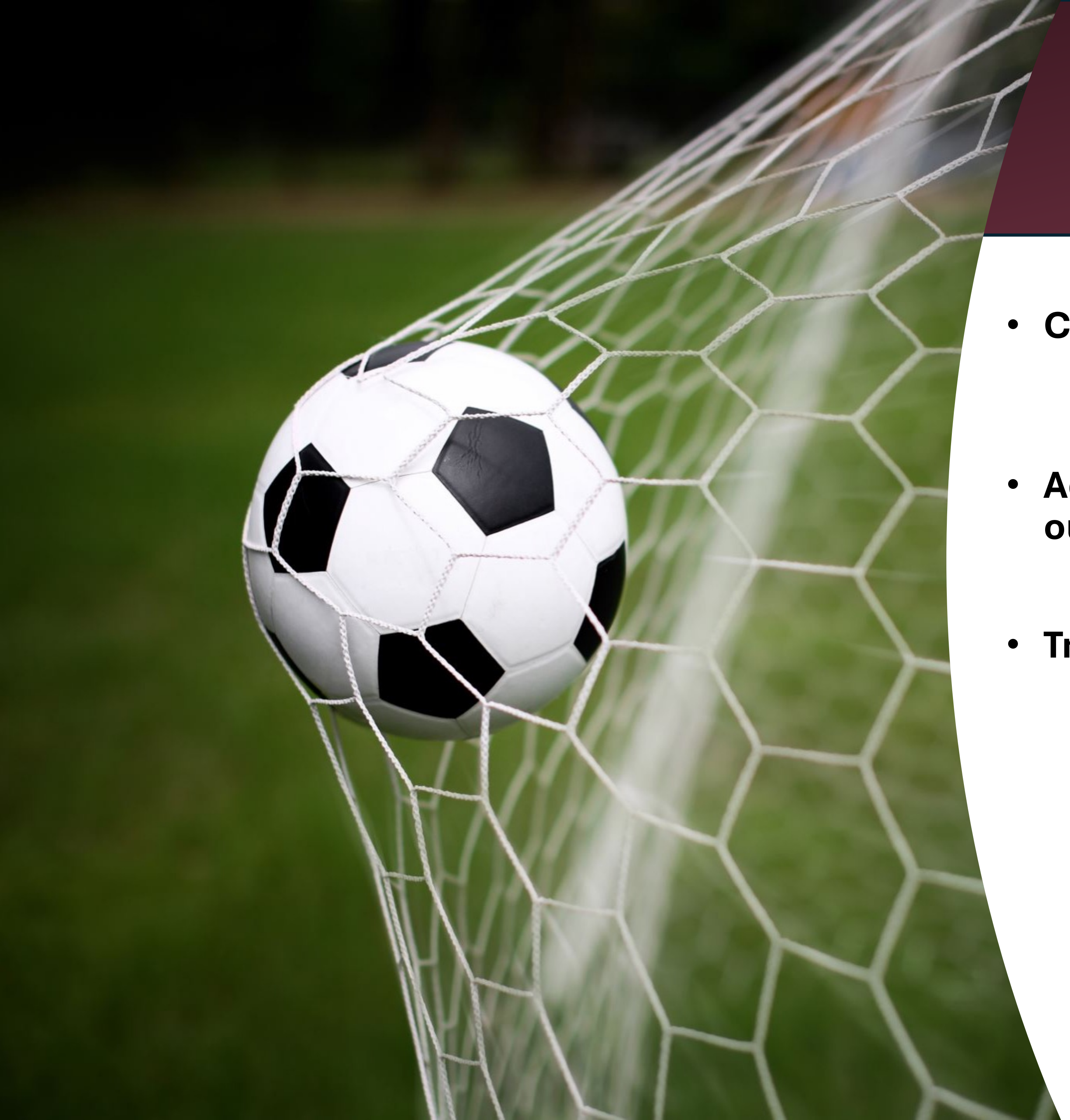
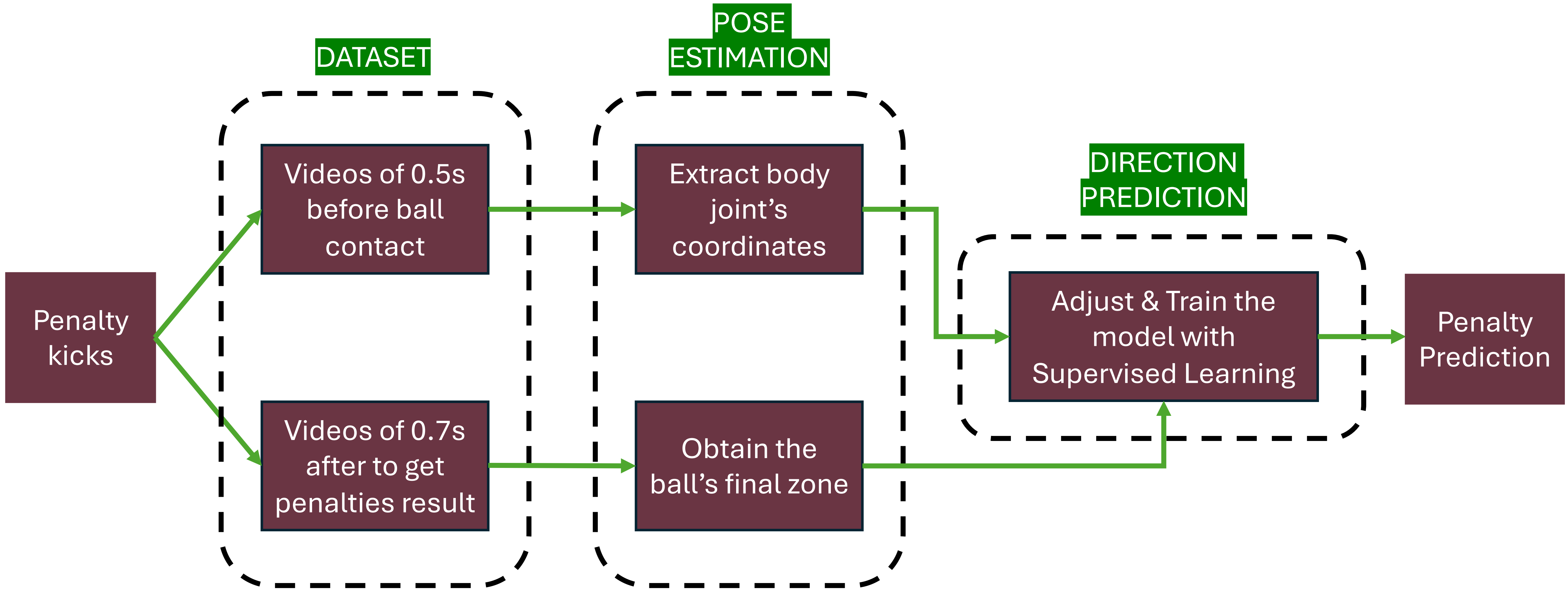


What’s next

- **Complete extraction of body joints coordinates**
  - Pre-trained model “*shufflenetv2k30*” from OpenPifPaf.
- **Adjust the existing model to obtain the correct output**
  - ST-Trans (UnPOSed github repository).
- **Train the model with supervised learning**
  - Feed the model with body joints coordinates from penalties videos.
  - Supervised learning with penalties result.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